Project: PollObs (no title for now)

Provisional author position:

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Abstract

Flowering plants exhibit an astonishing diversity of reproductive structures, which shapes how they interact with their pollinators. However, to what extent reproductive traits or other drivers like phenology and species abundances shape plant-pollinator interactions is still an open question. To disentangle the relative effects of species phenology, abundances and traits in plant-pollinator interactions, we compared pollinator visitation rates, diet breath, specialization from plant species that vary in reproductive traits, phenology and abundances. To do this, we observed plant-pollinator interactions for a whole flowering season in three different botanical gardens located in an urban setting. We anticipate that the ecological role of the different plan t and pollinator species are largely driven by their phenological overlap and abundance, and only certain taxonomic groups are strongly influenced by trait matching processes. Our results highlight the relevance of considering multiple mechanisms to understand plant-pollinator interactions.

Main objective

Explore how traits, phenology and abundances shape species interactions in three botanical gardens from Germany.

Pending tasks:

- Clean trait and phenology data for plants
- Get Pollinator phenology and traits
- Easy things to check: alpha, beta and gamma diversity of pollinators across the three gardens.

The data has an important temporal component

In Figure 1 we can see in a color coded NMDS the strong temporal component of the composition across gardens and different taxonomic levels.

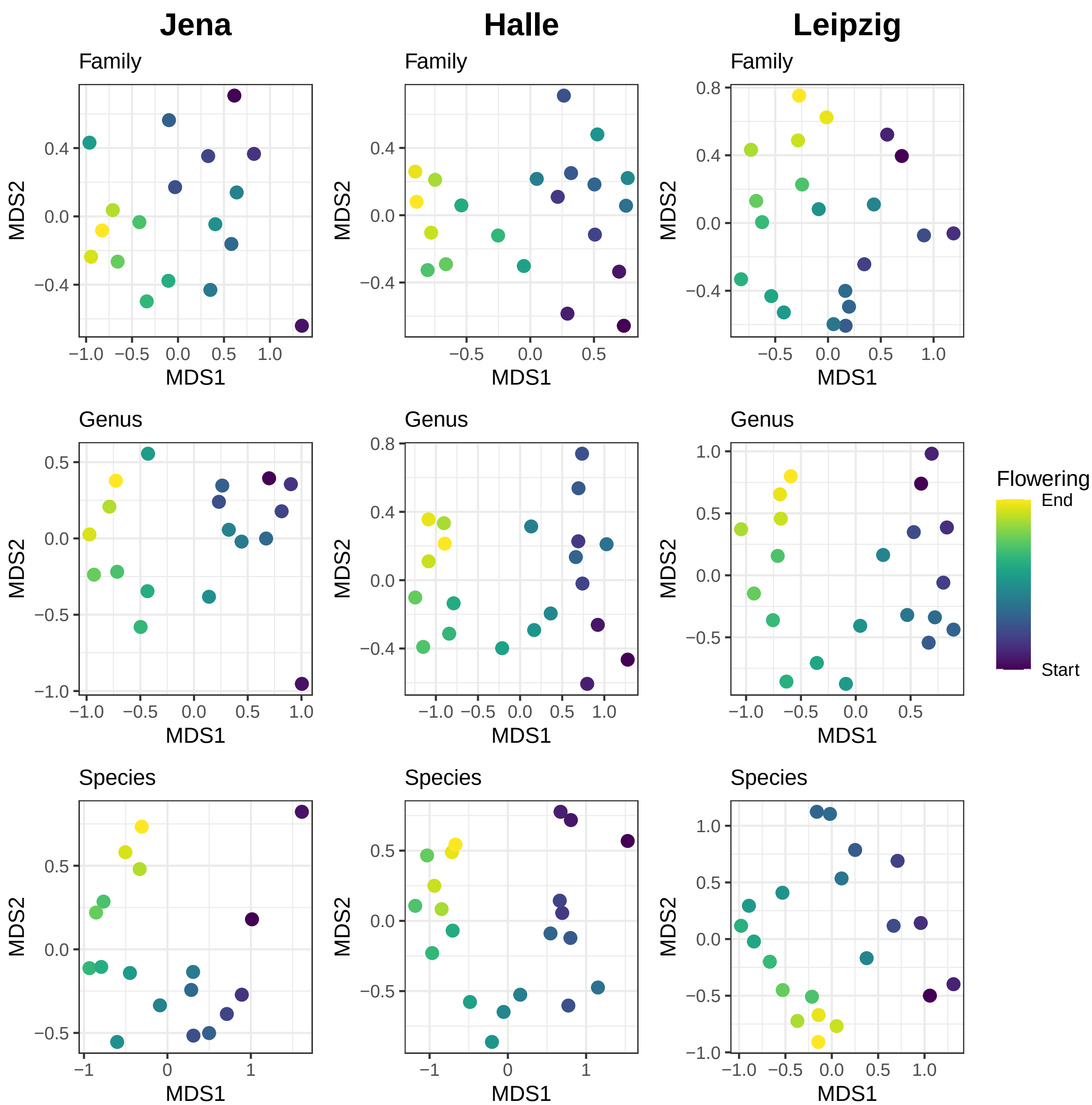
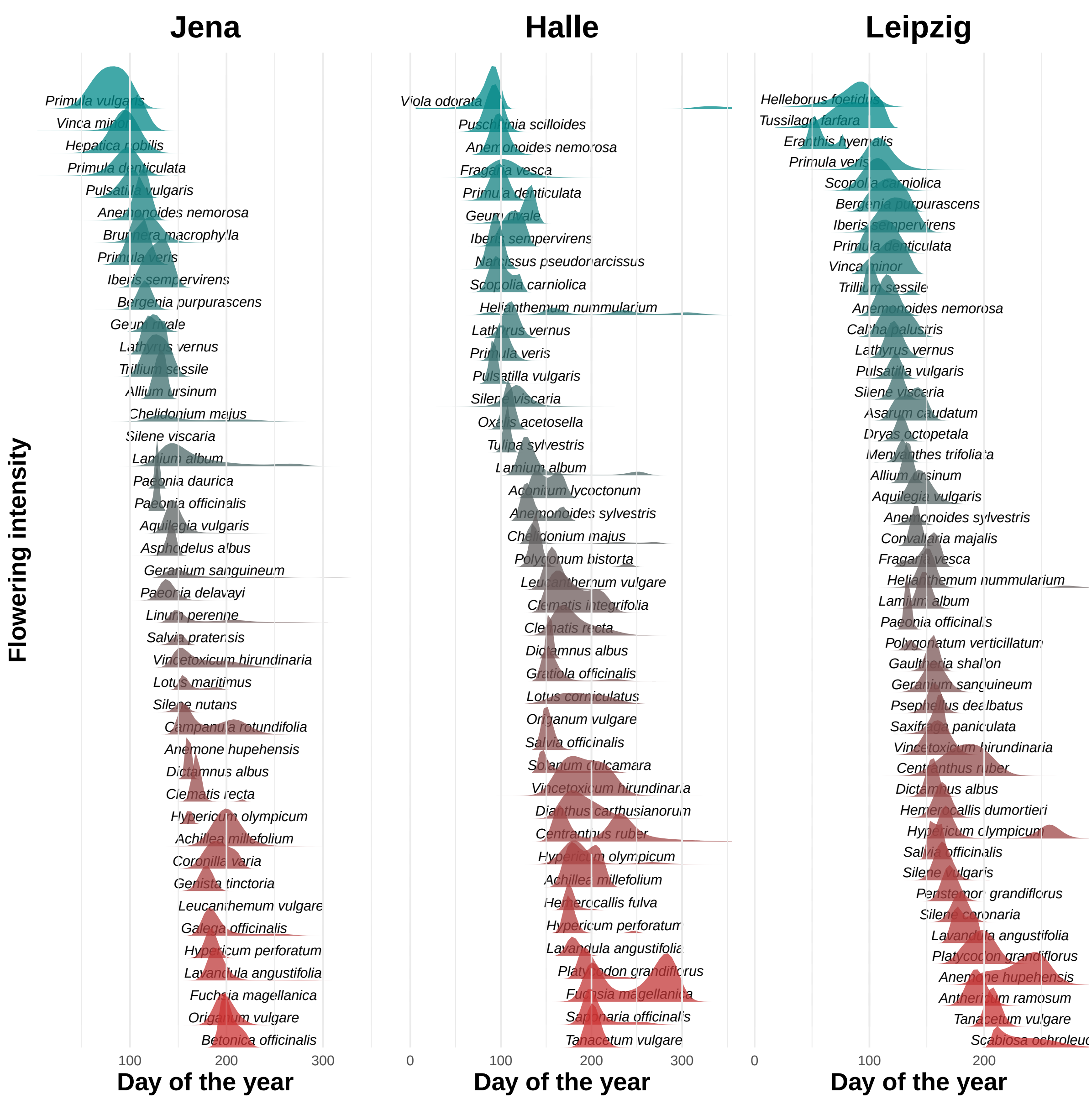


Figure 1. Non-metric multidimensional scaling (NMDS) of the pollinator community composition by sampling day in the three gardens at family, genus and species level.

Plant phenologies



Jena
Halle
Leipzig
Flowering intensity
Day of the year
Day of the year
Day of the year

Figure 2.